EXPERIMENT NO.3

DEVICE ONBOARDING AND DATA FLOW IN AWS IOT CORE (GOOGLE COLAB SIMULATION)

AIM

To design and implement a Smart Street Light Monitoring System by simulating device onboarding and data flow in AWS IoT Core using Python in Google Colab for monitoring the operational status of street lights.

OBJECTIVE

- To simulate the onboarding of smart street light devices.
- To register street light devices in AWS IoT Core.
- To authenticate each registered street light before data transmission.
- To simulate the transmission of street light status data (Working/Faulty).
- To monitor the operational status of street lights in real time.
- To identify faulty street lights requiring maintenance.
- To visualize the device onboarding status and monitoring results.
- To store the simulated data for future analysis.

SOFTWARE REQUIRED

- Google Colab
- Python 3
- Pandas
- Matplotlib
- Random Library

THEORY

AWS IoT Core is a cloud platform that enables secure communication between Internet of Things (IoT) devices and cloud applications. Before sending data to the cloud, every device must complete the **device onboarding** process, which includes registration and authentication.

In this experiment, a **Smart Street Light Monitoring System** is simulated using Python in Google Colab. Each street light is considered an IoT device. During onboarding, every street light is registered and authenticated with AWS IoT Core. After successful authentication, the device transmits its operational status, such as **Working** or **Faulty**, to the cloud.

The cloud receives the data, monitors the condition of all street lights, and identifies faulty lights that require maintenance. The collected information is stored for future analysis and visualized using graphs. This simulation demonstrates how AWS IoT Core can be used in smart city applications for efficient street light monitoring, remote management, and predictive maintenance.

SAMPLE DATA

Device ID	Street Light	Registration Status	Light Status
Device_001	L01	Registered	Working
Device_002	L02	Registered	Working
Device_003	L03	Registered	Faulty
Device_004	L04	Pending	Working
Device_005	L05	Registered	Working
Device_006	L06	Registered	Working
Device_007	L07	Registered	Faulty
Device_008	L08	Registered	Working
Device_009	L09	Registered	Working
Device_010	L10	Registered	Working

ALGORITHM

1. Start the program.
2. Initialize the total number of smart street light devices.
3. Generate a unique Device ID for each street light.
4. Register each device in the simulated AWS IoT Core environment.
5. Authenticate every registered device.
6. Generate the operational status of each street light (Working/Faulty).
7. Transmit the street light status data to the simulated AWS IoT Core.
8. Count the number of registered and pending devices.
9. Count the number of working and faulty street lights.

10. Display the Device Onboarding Summary.

11. Plot graphs to visualize the device status.

12. Store the monitoring data in a CSV file.

13. Stop the program.

OUTPUT

====== SMART STREET LIGHT MONITORING ======

Total Lights : 50

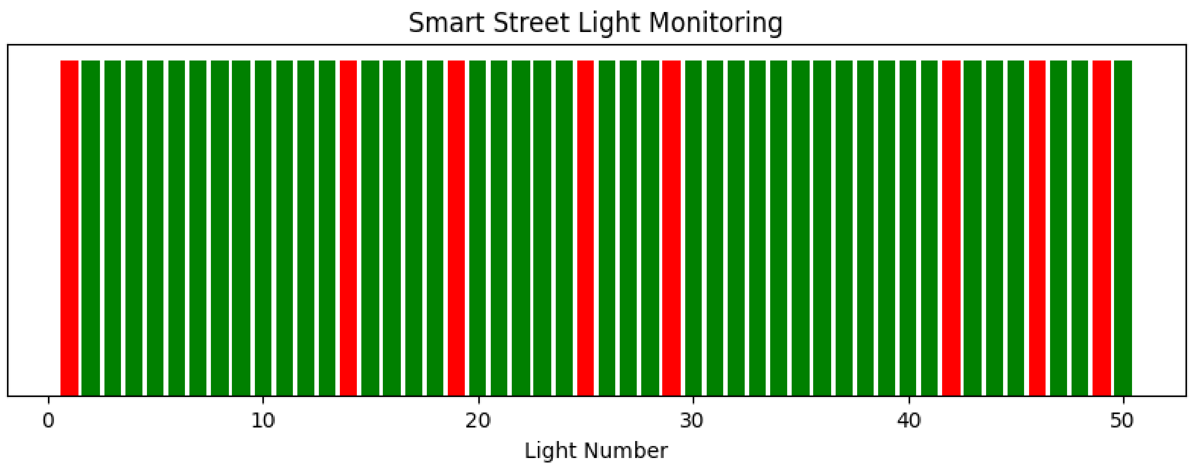
Working 42

Faulty 8

SMART STREET LIGHT MAP

💡 **= Working** + **= Faulty**

L01 +	**L02** 💡	**L03** 💡	**L04** 💡	**L05** 💡	**L06** 💡	**L07** 💡	**L08** 💡	**L09** 💡	**L10** 💡
L11 💡	**L12** 💡	**L13** 💡	**L14** +	**L15** 💡	**L16** 💡	**L17** 💡	**L18** 💡	**L19** +	**L20** 💡
L21 💡	**L22** 💡	**L23** 💡	**L24** 💡	**L25** +	**L26** 💡	**L27** 💡	**L28** 💡	**L29** +	**L30** 💡
L31 💡	**L32** 💡	**L33** 💡	**L34** 💡	**L35** 💡	**L36** 💡	**L37** 💡	**L38** 💡	**L39** 💡	**L40** 💡
L41 💡	**L42** +	**L43** 💡	**L44** 💡	**L45** 💡	**L46** +	**L47** 💡	**L48** 💡	**L49** +	**L50** 💡



RESULT : The **Device Onboarding and Data Flow in AWS IoT Core** simulation was successfully implemented using **Python in Google Colab**. The devices were registered, authenticated, and their data was transmitted to the simulated AWS IoT Core. The system successfully monitored the device status and demonstrated secure data flow between IoT devices and the cloud.